the end of the esophagus, the bolus encounters a **sphincter**, a ringlike valve of muscle **(Figure 41.10)**. Acting like a drawstring, the sphincter regulates passage of the ingested food into the next compartment, the stomach.

Digestion in the Stomach

The **stomach**, which is located just below the diaphragm, has two major roles in digestion. The first is storage. With a very elastic wall and accordion-like folds, the stomach can stretch to accommodate about 2 L of food and fluid. The second major function is to process food into a liquid suspension. The stomach secretes a digestive fluid called **gastric juice** and mixes it with the food through a churning action. This mixture of ingested food and gastric juice is called **chyme**.

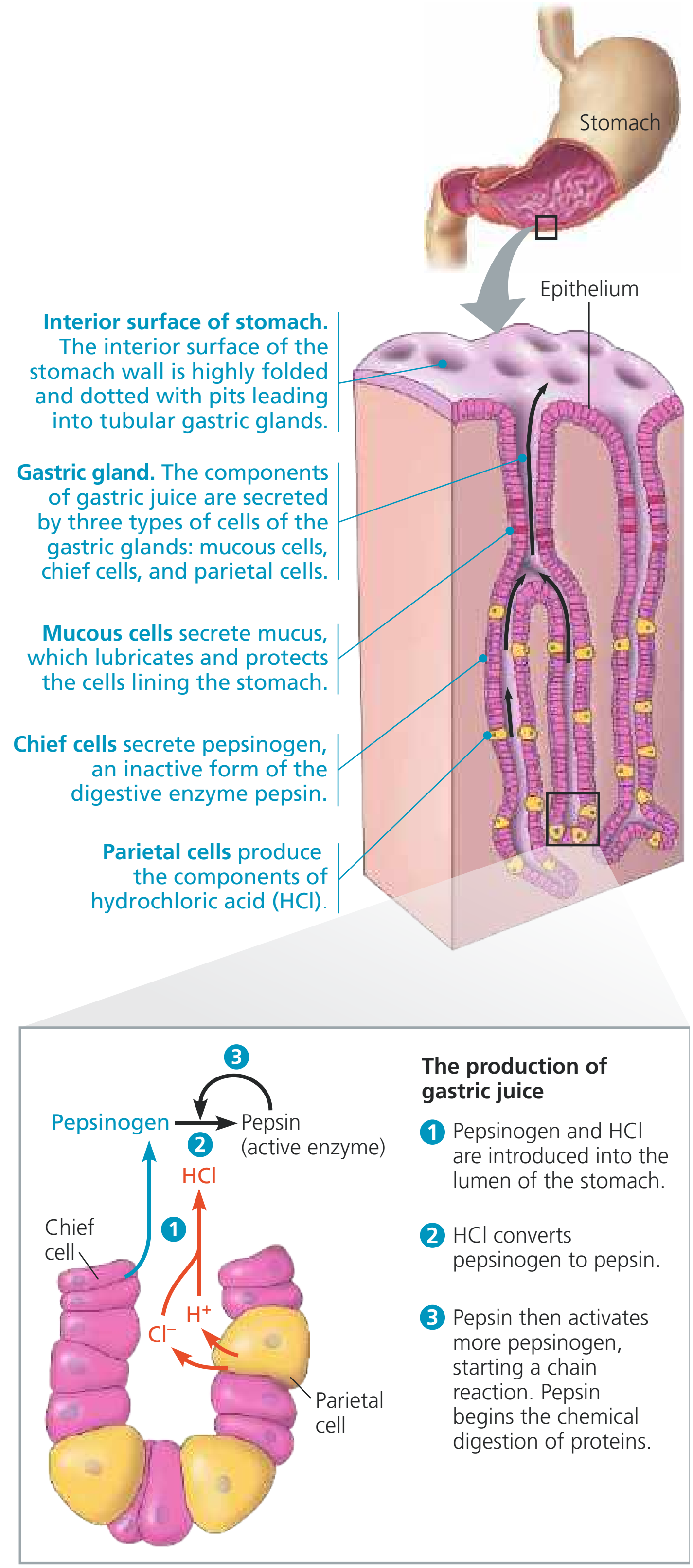
Chemical Digestion in the Stomach

Two components of gastric juice help liquefy food in the stomach. First, hydrochloric acid (HCl) disrupts the extracellular matrix that binds cells together in meat and plant material. The concentration of HCl is so high that the pH of gastric juice is about 2, acidic enough to dissolve iron nails (and to kill most bacteria). This low pH denatures (unfolds) proteins in food, increasing exposure of their peptide bonds. The exposed bonds are then attacked by the second component of gastric juice—a **protease**, or protein-digesting enzyme, called **pepsin**. Unlike most enzymes, pepsin is adapted to work best in a very acidic environment. By breaking peptide bonds, it cleaves proteins into smaller polypeptides and further exposes the contents of ingested tissues.

Two types of cells in the gastric glands of the stomach produce the components of gastric juice (see Figure 41.10). *Parietal cells* use an ATP-driven pump to expel hydrogen ions into the lumen. At the same time, chloride ions diffuse into the lumen through specific membrane channels of the parietal cells. It is therefore only within the lumen that hydrogen and chloride ions combine to form HCl. Meanwhile, *chief cells* release pepsin into the lumen in an inactive form called **pepsinogen**. HCl converts pepsinogen to active pepsin by clipping off a small portion of the molecule and exposing its active site. Through these processes, both HCl and pepsin form in the lumen (cavity) of the stomach, not within the cells of the gastric glands. As a result, the parietal and chief cells produce gastric juice but are not digested from within by its components.

After hydrochloric acid converts a small amount of pepsinogen to pepsin, pepsin itself helps activate the remaining pepsinogen. Pepsin, like HCl, can clip pepsinogen to expose the enzyme's active site. This generates more pepsin, which activates more pepsinogen. This series of events is an example of positive feedback (see Concept 40.2).

▼ Figure 41.10 The stomach and its secretions.



Mastering Biology **Figure Walkthrough**

Why don't HCl and pepsin eat through the lining of the stomach? For one thing, mucus secreted by cells in gastric glands protects against self-digestion (see Figure 41.10).